# 類別資料分析 第4章

**The Analysis of Categorical Data**

**Chapter 4 : Logistic Regression**


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# Logistic Regression

※Y : Binary response

X : Quantitative explanatory variable

$\pi(x) = P(Y = 1|X = x)$

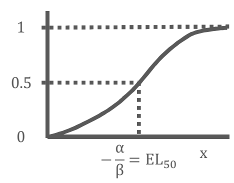
the logit regression model : $\text{logit}\left[\frac{\pi(x)}{1-\pi(x)}\right] = \alpha + \beta x \Rightarrow \pi(x) = \frac{e^{\alpha+\beta x}}{1+e^{\alpha+\beta x}} = \frac{1}{1+e^{-(\alpha+\beta x)}}, 0 < \pi(x) < 1$

(1) $\pi'(x) = \frac{d\pi(x)}{dx} = \frac{\beta e^{\alpha+\beta x}}{(1+e^{\alpha+\beta x})^2} = \beta \cdot \pi(x) \cdot [1 - \pi(x)]$

If $\pi(x) = 0.1$ , then $\pi'(x) = 0.09\beta$

If $\pi(x) = 0.9$ , then $\pi'(x) = 0.09\beta$

If $\pi(x) = 0.5$ , then $\pi'(x) = 0.25\beta$




當 $\pi(x) = 0.5$ 時，i. e. $x = -\frac{\alpha}{\beta}$ (is called the median effective level)， $\pi(x)$ 對 x 的圖形最陡峭

# Logistic Regression

(2) If $\beta = 0$ , then $\pi(x) = \frac{e^{\alpha}}{1+e^{\alpha}}$ , i. e. $\pi(x)$ is identical at all x $\Rightarrow$ The binary response Y is independent of X

(3) The interpretation of $\beta$ :

If $x = k + 1 \Rightarrow \ln\left[\frac{\pi(k+1)}{1-\pi(k+1)}\right] = \alpha + \beta \cdot (k+1)$

If $x = k \Rightarrow \ln\left[\frac{\pi(k)}{1-\pi(k)}\right] = \alpha + \beta \cdot k \Rightarrow \ln\frac{\pi(k+1)/[1-\pi(k+1)]}{\pi(k)/[1-\pi(k)]} = \beta$

$\therefore \theta = e^{\beta}$, where $\theta = \frac{\pi(k+1)/[1-\pi(k+1)]}{\pi(k)/[1-\pi(k)]} = \text{odds ratio}$

即 X 每增加一單位，$Y = 1$ 對 $Y = 0$ 的勝算為原來的 $e^{\beta}$ 倍

（即 X 每增加一單位，$Y = 1$ 對 $Y = 0$ 的勝算比原來的增加 $e^{\beta} - 1$ 倍）

# Logistic Regression

- 螃蟹例子改用 identity link :

  $X^2 = 3.0$ , $G^2 = 3.0$ , $df = 6$

  但利用 identity link 可能造成 predicted count 是負的

- 在評估某觀測值對 model fit 的影響時，可將該觀測值去除重新 fit

- Poisson 分配平均數和變異數相等，但在實務上變異數常比平均數大，此現象稱為 Over dispersion

# Logistic Regression

· 對多個解釋變數組合下的 response count 也許平均數和變異數相等，

但由於分析時只看某一種解釋變數，因此計算時將有相同 level 的值加總，

使得某一解釋變數組合下的 response count 過於分散。

(If the variance equals the mean when all relevant variables are controlled, it exceeds the mean when only one is controlled)

· 因為 Binomial 和 Poisson 分配，變異數為平均數的函數，因此較普遍有 over dispersion 的問題

· If $\mathrm{Var}(Y_i) \propto \mu_i$ , where $\mu_i = E(Y_i)$ ,

then $\frac{X^2}{df}$ estimate the proportionality constant $\Rightarrow$ adjusted ASE $= \sqrt{\frac{X^2}{df}} \cdot$ ASE

# Logistic Regression

e. g. 螃蟹追求者

$N = 66, X^2 = 174.3, df = 64, X^2/df = 2.7 \Rightarrow \sqrt{X^2/df} = \sqrt{174.3/64} = 1.65$

(1) $H_0 : \beta = 0$ vs. $\beta \neq 0$

(2) $\hat{\beta} = 0.164, \text{s.e.}(\hat{\beta}) = 0.02 \Rightarrow$ adjusted ASE $= 1.65 \cdot 0.002 = 0.033$

$\Rightarrow$ The adjusted Wald statistic $= \left(\frac{\hat{\beta}}{\text{adjusted ASE}}\right)^2 = \left(\frac{0.164}{0.033}\right)^2 \approx 24.8 \sim X_1^2$

(3) $\because$ p − value $< 0.0001 \therefore$ Reject $H_0$

· An over dispersion adjusted 95% C. I. for $\beta$ is $0.164 \pm 1.96 \cdot 0.033 = (0.099, 0.229)$

# Logistic Regression

※**The Newton-Raphson Algorithm**

L : the likelihood function, $L(\theta) = f(y;\theta)$ : the joint pdf of $y_1, y_2, \ldots, y_n$; $\ell(\theta) = \ln L(\theta)$

the normal equation : $\frac{\partial \ell(\theta)}{\partial \theta} = \begin{pmatrix} \frac{\partial \ell(\theta)}{\partial \theta_1} \\ \cdots \\ \frac{\partial \ell(\theta)}{\partial \theta_k} \end{pmatrix} = 0, \theta = (\theta_1, \ldots \theta_k)$

Assume $k = 1$, let $g(\theta) = \frac{\partial \ell(\theta)}{\partial \theta}$, 求 $g(\theta) = 0$ , $\theta = ?$

$\because$ 通過 $(\theta_0, g(\theta_0))$, 斜率為 $g'(\theta_0)$ 的直線方程式為 $y - g(\theta_0) = g'(\theta_0)(\theta - \theta_0)$,

$$\text{where } g'(\theta_0) = \left.\frac{\partial g(\theta)}{\partial \theta}\right|_{\theta=\theta_0}$$

$\therefore \theta_1 = \theta_0 - \frac{g(\theta_0)}{g'(\theta_0)} \Rightarrow \theta_{k+1} = \theta_k - \frac{g(\theta_k)}{g'(\theta_k)}, k = 1, 2, \ldots \ldots \Longrightarrow \theta_k$ converges to the root of $g(\theta) = 0$

# Logistic Regression

※**The Deviance**

(1) $\begin{cases} H_0 : \text{the fitted model} \\ H_1 : \text{the saturated model} \end{cases}$

(2) Deviance $= D^2 = -2 \ln \frac{\ell_M}{\ell_S} = -2(L_M - L_S) \sim X^2_{df}$

$df =$ the number of responses $-$ the number of parameters in the fitted model

(3) $D^2$ 要小表示所選擇的模式 fit 好

Reject $H_0$ if $D^2 > X^2_{df,\alpha}$

# Logistic Regression

※The Deviance

e. g. $\begin{cases} H_0 : \ln\frac{\pi(x)}{1-\pi(x)} = \alpha \\ H_1 : \ln\frac{\pi(x)}{1-\pi(x)} = \alpha + \beta x \end{cases} \Rightarrow \begin{cases} \pi_0(x) = \frac{e^{\alpha}}{1+e^{\alpha}} \\ \pi_1(x) = \frac{e^{\alpha+\beta x}}{1+e^{\alpha+\beta x}} \end{cases}$

| | | Y | | |
|---|---|---|---|---|
| | | 1 | 2 | |
| X | 0 | $n_{11}$ | $n_{12}$ | $n_{1+}$ |
| | 1 | $n_{21}$ | $n_{22}$ | $n_{2+}$ |

$n_{1+}$ and $n_{2+}$ are fixed

$n_{11} \sim Bin(n_{1+}, \pi(0))$

$n_{21} \sim Bin(n_{2+}, \pi(1))$

$$\ell_M = \binom{n_{1+}}{n_{11}} [\hat{\pi}_0(0)]^{n_{11}} [1-\hat{\pi}_0(0)]^{n_{12}} \binom{n_{2+}}{n_{21}} [\hat{\pi}_0(1)]^{n_{21}} [1-\hat{\pi}_0(1)]^{n_{22}}$$

where $\hat{\pi}_0(0) = \frac{e^{\hat{\alpha}_0}}{1+e^{\hat{\alpha}_0}} = \hat{\pi}_0(1)$, $\hat{\alpha}_0$ : MLE under $H_0$

$$\ell_s = \binom{n_{1+}}{n_{11}} [\hat{\pi}_1(0)]^{n_{11}} [1-\hat{\pi}_1(0)]^{n_{12}} \binom{n_{2+}}{n_{21}} [\hat{\pi}_1(1)]^{n_{21}} [1-\hat{\pi}_1(1)]^{n_{22}}$$

where $\hat{\pi}_1(0) = \frac{e^{\hat{\alpha}_1}}{1+e^{\hat{\alpha}_1}}, \hat{\pi}_1(1) = \frac{e^{\hat{\alpha}_1+\hat{\beta}_1 x}}{1+e^{\hat{\alpha}_1+\hat{\beta}_1 x}}, \hat{\alpha}_1, \hat{\beta}_1$: MLE under $H_1$

# Logistic Regression

· For 2 × 2 table, under $H_1$ : $\ln\frac{\pi(x)}{1-\pi(x)} = \alpha + \beta x$ is a saturated model

$\because \begin{cases} \ln\frac{n_{11}}{n_{12}} = \alpha \\ \ln\frac{n_{21}}{n_{22}} = \alpha + \beta x \end{cases} \Rightarrow \begin{cases} \hat{\alpha} = \ln\frac{n_{11}}{n_{12}} \\ \hat{\beta} = \ln\frac{n_{11}/n_{12}}{n_{21}/n_{22}} = \ln\hat{\theta}, \hat{\theta} \text{ : sample odds ratio} \end{cases}$

· Under $H_0$ : $\pi(x) = \frac{e^{\alpha}}{1+e^{\alpha}} \Rightarrow \pi_0(0) = \pi_0(1) = \frac{e^{\alpha}}{1+e^{\alpha}}$

$$\Rightarrow \ell_M \propto \left(\frac{e^{\alpha}}{1+e^{\alpha}}\right)^{n_{11}+n_{21}} \left(1-\frac{e^{\alpha}}{1+e^{\alpha}}\right)^{n_{12}+n_{22}}$$

$$\text{i.e. } L_M \propto (n_{11}+n_{21})\ln\left(\frac{e^{\alpha}}{1+e^{\alpha}}\right) + (n_{12}+n_{22})\ln\left(1-\frac{e^{\alpha}}{1+e^{\alpha}}\right)$$

$$\propto \alpha(n_{11}+n_{21}) - (n_{11}+n_{21}+n_{12}+n_{22})\ln(1+e^{\alpha})$$

# Logistic Regression

$$\Rightarrow \frac{\partial L_M}{\partial \alpha} = (n_{11} + n_{21}) - \frac{e^{\alpha}}{1+e^{\alpha}} \cdot (n_{11} + n_{21} + n_{12} + n_{22}) = 0$$

$$\therefore \frac{e^{\alpha}}{1+e^{\alpha}} = \frac{n_{11}+n_{21}}{n_{11}+n_{21}+n_{12}+n_{22}} \Rightarrow \hat{\alpha} = \ln \frac{\hat{\pi}_0(1)}{1-\hat{\pi}_0(1)}, \hat{\pi}_0(1) = \frac{n_{11}+n_{21}}{n_{11}+n_{21}+n_{12}+n_{22}} = \frac{n_{+1}}{n_{++}} = \hat{\pi}_0(0)$$

# Logistic Regression

※可利用 $\Delta D^2$ 來檢定 2 個 nested model，i. e . Model 0 是 Model 1 的 reduce model。例如：

$$\begin{cases} H_0: \ln \mu = \alpha + \beta_1 x_1 \quad (\text{model } 0) \\ H_1: \ln \mu = \alpha + \beta_1 x_1 + \beta_2 x_2 (\text{model } 1) \end{cases} \Leftrightarrow \begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$$

Under $H_0$ , $\ell_0 \propto \prod_i e^{-\hat{\mu}_i^{(0)}} \cdot \left[\hat{\mu}_i^{(0)}\right]^{y_i}, \hat{\mu}_i^{(0)}$ : MLE of $\mu_i$ for model 0

Under $H_0$ , $\ell_1 \propto \prod_i e^{-\hat{\mu}_i^{(1)}} \cdot \left[\hat{\mu}_i^{(1)}\right]^{y_i}, \hat{\mu}_i^{(1)}$ : MLE of $\mu_i$ for model 1

$$\Rightarrow -2 \ln \frac{\ell_0}{\ell_1} = -2(L_0 - L_1) = \Delta D^2 \sim X^2_{df},$$

df = the number of parameters in model 1 − the number of parameters in model 0

# Logistic Regression

· 將此檢定分 2 部分來看：

$\begin{cases} H_0: \ln\mu = \alpha + \beta_1 x_1 \ (\text{model } 0) \\ H_1: \text{Saturated model} \end{cases}$ $\begin{cases} H_0: \ln\mu = \alpha + \beta_1 x_1 + \beta_2 x_2 (\text{model } 1) \\ H_1: \text{Saturated model} \end{cases}$

$\Rightarrow -2(L_0 - L_S) = D_0^2 = -2\ln\frac{\ell_0}{\ell_s} \sim X^2_{df(0)}$ ; $-2(L_1 - L_S) = D_1^2 = -2\ln\frac{\ell_1}{\ell_s} \sim X^2_{df(1)}$

where $\ell_s \propto \prod_i e^{-\hat{\mu}_i^{(s)}} \cdot \left[\hat{\mu}_i^{(s)}\right]^{y_i}$, $\hat{\mu}_i^{(s)}$ : MLE of $\mu_i$ for saturated model

$df(0) = n -$ the number of parameters in model 0 ;

$df(1) = n -$ the number of parameters in model 1

$\Rightarrow \Delta D^2 = D_0^2 - D_1^2 = -2\ln\frac{\ell_0}{\ell_s} + 2\ln\frac{\ell_1}{\ell_s} = -2\ln\frac{\ell_0}{\ell_1} \approx X^2_{df}$ 當 df 少時，即使 $\mu_i \le 5$，卡方近似仍非常好

df $=$ the number of parameters in model 1 $-$ the number of parameters in model 0 $= df(0) - df(1)$

# Logistic Regression

e. g. 螃蟹追求者 (fit the logistic regression)

$\hat{\alpha} = -12.351, \hat{\beta} = 0.497, Y_i = 1$ if $(Satellites)_i \ge 1$

As $x = 26.3$ : $\hat{\pi}(x) = \frac{e^{-12.351+0.497\cdot 26.3}}{1+e^{-12.351+0.497\cdot 26.3}} = 0.674$ : predicted probability

$\Rightarrow \widehat{odds}_1 = \frac{\hat{\pi}(x)}{1-\hat{\pi}(x)} = \frac{0.674}{0.326} \approx 2.07$

As $x = 27.3$ : $\hat{\pi}(x+1) = \frac{e^{-12.351+0.497\cdot 27.3}}{1+e^{-12.351+0.497\cdot 27.3}} = 0.773$

$\Rightarrow \widehat{odds}_2 = \frac{\hat{\pi}(x+1)}{1-\hat{\pi}(x+1)} = \frac{0.773}{0.227} \approx 3.40 \Longrightarrow \frac{\widehat{odds}_2}{\widehat{odds}_1} = \frac{3.40}{2.07} = 1.64 = e^{\hat{\beta}} = e^{0.497}$

∴ 當蟹殼寬度增加 1 公分，螃蟹的追求者至少有 1 隻對沒有追求者的勝算增加 64%

# Logistic Regression

※The predicted probability : $\hat{\pi}(x) = \frac{e^{\hat{\alpha}+\hat{\beta}x}}{1+e^{\hat{\alpha}+\hat{\beta}x}}$

In Figure 4.1, the sample proportion $= \frac{\text{the number of having satellites}}{\text{the number of cases}}$

e. g. $\frac{\text{the number of } (X_i<23.25 \text{ and } Y_i=1)}{\text{the number of } X_i<23.25} = \frac{5}{14} = 0.36$

· Predicted number of crabs = the number of cases × predicted probability

As $x = 22.69$ , $\hat{\pi}(x) = \frac{e^{-12.351+0.497\cdot 22.69}}{1+e^{-12.351+0.497\cdot 22.69}} = 0.26$

$\Rightarrow$ predicted number of crabs $= 14 \times 0.26 = 3.64$

As $x = -\frac{\hat{\alpha}}{\hat{\beta}} = \frac{12.351}{0.487} = 24.8$ , $\hat{\pi}(x) = 0.5$

As $x = 33.5$ , $\hat{\pi}(x) = 0.987$

# Logistic Regression

· Suppose $X|Y = 1 \sim N(\mu_1, \sigma^2)$ , $f_1(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu_1)^2}{2\sigma^2}}$

$X|Y = 0 \sim N(\mu_0, \sigma^2)$ , $f_0(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu_0)^2}{2\sigma^2}}$

$\Rightarrow f_X(x) = p \cdot f_1(x) + (1-p) \cdot f_0(x)$, where $p = P(Y = 1)$

$\pi(x) = P(Y = 1|X = x) = \frac{P(x \le X \le x+dx, Y=1)}{P(x \le X \le x+dx)} = \frac{p\cdot f_1(x)}{p\cdot f_1(x)+(1-p)\cdot f_0(x)}$

$\frac{\pi(x)}{1-\pi(x)} = \frac{p\cdot f_1(x)}{(1-p)\cdot f_0(x)} = \frac{p}{1-p} e^{\frac{1}{2\sigma^2}[(x-\mu_0)^2-(x-\mu_1)^2]} = \frac{p}{1-p} e^{\frac{1}{2\sigma^2}[2(\mu_1-\mu_0)+\mu_0^2-\mu_1^2]}$

$\Rightarrow \ln \frac{\pi(x)}{1-\pi(x)} = \frac{1}{\sigma^2}(\mu_1 - \mu_0)x + \frac{\mu_0^2-\mu_1^2}{2\sigma^2} + \ln \frac{p}{1-p}$

$\Rightarrow$ 故 $\beta$ 和 $\mu_1 - \mu_0$ 有相同符號

$\beta = \frac{1}{\sigma^2}(\mu_1 - \mu_0)$ , $\alpha = \frac{\mu_0^2-\mu_1^2}{2\sigma^2} + \ln \frac{p}{1-p}$

# Logistic Regression

※Confidence Interval of $\beta$ : $\hat{\beta} \pm Z_{\alpha/2} \cdot \text{ASE}$, $\text{ASE} = [\text{In}(\beta)]^{-\frac{1}{2}}$

$\because$ $\hat{\beta}$ is the MLE of $\beta$

$\therefore$ $\hat{\beta} \sim N(\beta, \sigma^{2(\hat{\beta})})$, where $\sigma^2(\hat{\beta}) = [\text{In}(\beta)]^{-1}$, $\text{In}(\beta) = -E\left[\frac{\partial^2 \ln \ell(\alpha,\beta)}{\partial \beta^2}\right]$ : the Fisher's Information

the likelihood function $\ell(\alpha, \beta) = \prod_{i=1}^{n} [\pi(x_i)]^{y_i} [1 - \pi(x_i)]^{1-y_i}$

$\pi(x_i) = \frac{e^{\alpha+\beta x}}{1+e^{\alpha+\beta x}} \left(\because \ln \frac{\pi(x_i)}{1-\pi(x_i)} = \alpha + \beta \cdot x_i\right)$ $\because Y_i|X_i = x_i \sim \text{Ber}(\pi(x_i))$

# Logistic Regression

e. g. 螃蟹追求者例子

$\hat{\beta} = 0.497, \text{ASE} = 0.102$

$\Rightarrow$ A 95% C. I. for $\beta$ is $0.497 \pm 1.96 \cdot 0.102 = (0.298, 0.697)$

A 95% C. I. for the effect on the odds per centimeter increase in width x

equals $(e^{0.298}, e^{0.697}) = (1.35, 2.01)$

$\because$ $\pi'(x) = \beta \cdot \pi(x) \cdot [1 - \pi(x)]$ : approximates the change in the probability per unit change in x

e. g. $\pi(x) = 0.5 \Rightarrow \pi'(x) = 0.25\beta$

$\Rightarrow$ A 95% C. I. for $0.25\beta$ is $(0.25 \times 0.298, 0.25 \times 0.697) = (0.074, 0.174)$

即蟹殼寬度每增加 1 公分，有追求者機率增加速率$\pi'(x)$介於 0.07 和 0.17 之間。

# Logistic Regression

※Significance Testing

(1) Wald test

1. $\begin{cases} H_0: \beta = 0 \\ H_1: \ln\frac{\pi(x)}{1-\pi(x)} = \alpha + \beta x \end{cases}$

2. $Z = \frac{\hat{\beta}}{ASE} \sim N(0,1) \Rightarrow$ Wald statistic $Z^2 \sim X_1^2$

3. Reject $H_0$ if $Z^2 > X_{1,\alpha}^2$

# Logistic Regression

※Significance Testing

(2) Likelihood-ratio test（通常概似比檢定較華德檢定更具檢定力）

1. $\begin{cases} H_0: \beta = 0 \\ H_1: \ln\frac{\pi(x)}{1-\pi(x)} = \alpha + \beta x \end{cases}$

2. $-2\log(\ell_0 - \ell_1) = -2(L_0 - L_1) \sim X_1^2$ ,

where $\ell_0 = \ell(\hat{\alpha}_0) = \prod_{i=1}^{n}\left(\frac{e^{\hat{\alpha}_0}}{1+e^{\hat{\alpha}_0}}\right)^{y_i}\left(1-\frac{e^{\hat{\alpha}_0}}{1+e^{\hat{\alpha}_0}}\right)^{1-y_i}$, $\hat{\alpha}_0$ : MLE of $\alpha$ under $H_0$

$\ell_1 = \ell(\hat{\alpha}, \hat{\beta}) = \prod_{i=1}^{n}\left(\frac{e^{\hat{\alpha}+\hat{\beta}x}}{1+e^{\hat{\alpha}+\hat{\beta}x}}\right)^{y_i}\left(1-\frac{e^{\hat{\alpha}+\hat{\beta}x}}{1+e^{\hat{\alpha}+\hat{\beta}x}}\right)^{1-y_i}$, $\hat{\alpha}, \hat{\beta}$: MLE of $\alpha$ and $\beta$ under $H_1$

3. Reject $H_0$ if $-2(L_0 - L_1) > X_{1,\alpha}^2$

# Logistic Regression

e. g. 螃蟹追求者例子

**(1) Wald test**

$$\mathbf{Z = \frac{\hat{\beta}}{ASE} = \frac{0.497}{0.102} = 4.9 \Rightarrow Z^2 = 23.9 > X^2_{1,0.05} = 3.84}$$

**(2) Likelihood-ratio test**

$\mathbf{L_0 = -112.88}$ **under** $\mathbf{H_0 : \beta = 0\ ;\ L_1 = -97.23}$

$\mathbf{\Rightarrow -2(L_0 - L_1) = 31.3 > X^2_{1,0.05} = 3.84}$

# Logistic Regression

**※Distribution of probability estimates**

$$\mathbf{\pi(x) = P(Y = 1|X = x) = \frac{e^{\alpha+\beta x}}{1+e^{\alpha+\beta x}} \Rightarrow \hat{\pi}(x) = \hat{P}(Y = 1|X = x) = \frac{e^{\hat{\alpha}+\hat{\beta} x}}{1+e^{\hat{\alpha}+\hat{\beta} x}}}$$

**How to find the** $\mathbf{100(1-\alpha)\%}$ **C. I. for** $\mathbf{\pi(x)}$ **?**

· **A** $\mathbf{100(1-\alpha)\%}$ **C. I. of** $\mathbf{\alpha + \beta x}$ **is** $\mathbf{(\hat{\alpha} + \hat{\beta}x) + Z_{\alpha/2}\left[\widehat{Var}\left(\hat{\alpha} + \hat{\beta}x\right)\right]^{\frac{1}{2}}}$

$\mathbf{Var\left(\hat{\alpha} + \hat{\beta}x\right) = Var(\hat{\alpha}) + x^2 \cdot Var\left(\hat{\beta}\right) + 2 \cdot x \cdot Cov\left(\hat{\alpha}, \hat{\beta}\right)}$

# Logistic Regression

※Distribution of probability estimates

Let $\theta = (\alpha, \beta), \hat{\theta} = (\hat{\alpha}, \hat{\beta})$ : MLE of $(\alpha, \beta)$

$\Rightarrow$ (1) $\hat{\alpha}$ is a consistent estimator of $\alpha$ (i. e. $P(|\hat{\alpha} - \alpha| > \epsilon) \to 0$)

$\hat{\beta}$ is a consistent estimator of $\beta$ (i. e. $P(|\hat{\beta} - \beta| > \epsilon) \to 0$)

(2) $\hat{\theta} \sim N(\theta, I(\theta)^{-1})$

where $I(\theta)^{-1} = \begin{pmatrix} Var(\hat{\alpha}) & Cov(\hat{\alpha}, \hat{\beta}) \\ Cov(\hat{\alpha}, \hat{\beta}) & Var(\hat{\beta}) \end{pmatrix}, I(\theta) = \begin{pmatrix} \frac{\partial^2 \ln \ell(\alpha,\beta)}{\partial \alpha^2} & \frac{\partial^2 \ln \ell(\alpha,\beta)}{\partial \alpha\, \partial \beta} \\ \frac{\partial^2 \ln \ell(\alpha,\beta)}{\partial \alpha\, \partial \beta} & \frac{\partial^2 \ln \ell(\alpha,\beta)}{\partial \beta^2} \end{pmatrix}$

# Logistic Regression

e. g. 螃蟹追求者 : $\hat{\alpha} = -12.351$ , $\hat{\beta} = 0.497$

$x = 26.5 \Rightarrow \hat{\pi}(x) = \frac{e^{-12.351+0.497\cdot 26.5}}{1+e^{-12.351+0.497\cdot 26.5}} = 0.695$

$\hat{\alpha} + \hat{\beta}x = -12.351 + 0.497 \cdot 26.5 = 0.825$

$\widehat{Var}(\hat{\alpha}) = 6.91$ , $\widehat{Var}(\hat{\beta}) = 0.01035$ , $\widehat{Cov}(\hat{\alpha}, \hat{\beta}) = -0.02668$

$\Rightarrow \widehat{Var}(\hat{\alpha} + \hat{\beta}x) = 6.91 + 26.5^2 \cdot 0.01035 + 2 \cdot 26.5 \cdot (-0.2668) = 0.038$

$\therefore$ The 95% C. I. for $\alpha + \beta x$ is $0.825 \pm 1.96 \cdot \sqrt{0.038} = (0.44, 1.21)$

$\Rightarrow$ The 95% C. I. for $\pi(x)$ is $\left(\frac{e^{0.44}}{1+e^{0.44}}, \frac{e^{1.21}}{1+e^{1.21}}\right) = (0.61, 0.77)$

# Logistic Regression

e. g. 螃蟹追求者 : $\hat{\alpha} = -12.351$ , $\hat{\beta} = 0.497$

The sample proportion estimate at $x = 26.5$ is $\hat{P} = \frac{4}{6} = 0.67$

A $100(1-\alpha)\%$ C. I. of $P(Y = 1|X = 26.5)$ is $\hat{P} \pm Z_{\alpha/2}\sqrt{\hat{P}(1-\hat{P})/n}$

$\Rightarrow$ A 95% C. I. of P is $0.67 \pm 1.96 \cdot \sqrt{0.67 \cdot 0.33/6} = (0.22, 0.96)$

· C. I. =$(0.61, 0.77)$ 是經由 model 建立 Y 和 X 的關係式所得區間，C. I. =$(0.22, 0.96)$ 並無建立 model

當 model 正確時，C. I. =$(0.61, 0.77)$ 較 $(0.22, 0.96)$ 區間長度短，且估計較準確。

但事實上，任何模式並不能完全反應 $\pi(x)$ 和 x 間的關係，

因此，經由 model 所得的 C. I. 可能當樣本數增加時，仍無法收斂至母體參數。

# Logistic Regression

※Model checking

(1) Chi-square goodness of fit

1. $\begin{cases} H_0: \text{fitted model} \quad (\text{e.g. logit } \pi(x) = \alpha + \beta x) \\ H_1: \text{saturated model} \end{cases}$

2. $X^2 = \sum \frac{(\text{observed}-\text{fitted})^2}{\text{fitted}} \sim \chi^2_{df}$

$df =$ the number of different categories of explanatory variables

$-$ the number of parameters in fitted model

3. Reject $H_0$ if $X^2 > \chi^2_{df,\alpha}$

# Logistic Regression

※Model checking

$\pi(x) = P(Y = 1|x) = \frac{e^{\alpha+\beta_1 x_1}}{1+e^{\alpha+\beta_1 x_1}}$

Y : 有無肺癌

$X_1$ : 性別

$X_2$ : 有無抽菸

(2) Likelihood-ratio goodness of fit

$$1. \begin{cases} H_0 : \text{fitted model} & \left(\log\frac{\pi(x)}{1-\pi(x)} = \alpha + \beta_1 x_1\right) \\ H_1 : \text{saturated model} & \left(\log\frac{\pi(x)}{1-\pi(x)} = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2\right) \end{cases}$$

2. $\text{Deviance} = -2\ln\lambda = -2\ln\frac{\ell_0}{\ell_1} = -2(L_0 - L_1) \sim \chi^2_{df}$

$\ell_0$ : the maximum likelihood function under $H_0$

$\ell_1$ : the maximum likelihood function under $H_1$

3. Reject $H_0$ if $X^2 > \chi^2_{df,\alpha}$

| $X_1$ : 男 | | Y: 1 | Y: 2 |
|---|---|---|---|
| $X_2$ | 0 | | |
| | 1 | | |

| $X_1$ : 女 | | Y: 1 | Y: 2 |
|---|---|---|---|
| $X_2$ | 0 | | |
| | 1 | | |

Saturated model for $2 \times 2 \times 2$ contingency tables

# Logistic Regression

※Model checking

- $X^2$ and $G^2$ are applied for a fixed number of cells when the fitted counts are large.
- $X^2$ and $G^2$ for logistic regression models fitted with continuous or nearly-continuous predictors do not have approximate chi-squared distribution.

# Logistic Regression

e. g. 螃蟹追求者 : $\hat{\alpha} = -12.351$ , $\hat{\beta} = 0.497$

$$\begin{cases} H_0 : \text{logit } \pi(x) = \alpha + \beta x \\ H_1 : \text{saturated model} \end{cases}$$

· fitted value = the number of cases × predicted probability

| | | X : width | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | $< 23.25$ | $23.25 - 24.25$ | $24.25 - 25.25$ | $25.25 - 26.25$ | $26.25 - 27.25$ | $27.25 - 28.25$ | $28.25 - 29.25$ | $> 29.25$ |
| Y | Yes | 5<br>3.64 | 4<br>5.31 | 17<br>13.78 | 21<br>24.23 | 15<br>15.94 | 20<br>19.38 | 15<br>15.65 | 14<br>13.08 |
| | No | 9<br>10.36 | 10<br>8.69 | 11<br>14.77 | 18<br>14.77 | 7<br>6.06 | 4<br>4.62 | 3<br>2.35 | 0<br>0.92 |
| | | 14 | 14 | 28 | 39 | 22 | 24 | 18 | 14 |

observed

fitted

the number of cases

$$X^2 = \frac{(5-3.64)^2}{3.64} + \frac{(4-5.31)^2}{5.31} + \cdots + \frac{(0-0.92)^2}{0.92} = 5.3 < X^2_{6,0.05} ,$$

$$G^2 = -2(L_0 - L_1) = 6.2 < X^2_{6,0.05} , \text{df} = 8 - 2 = 6$$

# Logistic Regression

e. g. 螃蟹追求者 : $\hat{\alpha} = -12.351$ , $\hat{\beta} = 0.497$

∴ 母蟹是否有追求者的機率只與蟹殼寬度有關的 logit model fit well

· 故當解釋變數為連續隨機變數時，需適當分類後再作 goodness of fit test

· An alternative way of grouping forms observed and fitted values based on a partitioning of predicted probabilities. e. g. To form 10 groups, each pair of observed and fitted counts refers to $\frac{n}{10}$ observations.

· For each group, the fitted value for an outcome is the sum of the predicted probabilities for that outcome for all observations in that group $X^2 \sim X^2_{g-2}$ , g : the number of groups.

# Logistic Regression

· Likelihood-ratio test

1. $\begin{cases} H_0: \text{model } 0 \\ H_1: \text{model } 1 \end{cases}$ $\left(\text{e.g.} \begin{matrix} H_0: \text{logit } \pi(x) = \alpha + \beta_1 x_1 \\ H_1: \text{logit } \pi(x) = \alpha + \beta_1 x_1 + \beta_2 x_2 \end{matrix}\right)$

2. $\Delta G^2 = G^2(M_0|M_1) = -2(L_0 - L_1) = -2(L_0 - L_S) - [-2(L_1 - L_S)] = G^2(M_0) - G^2(M_1) \sim X^2_{df}$

$df = \text{the number of parameters in } H_1 - \text{the number of parameters in } H_0$

$= \text{df of } G^2(M_0) - \text{df of } G^2(M_1) = df(0) - df(1)$

$\begin{cases} H_0: \text{model } 0 \\ H_1: \text{Saturated model} \end{cases}$ $\begin{cases} H_0: \text{model } 1 \\ H_1: \text{Saturated } \textit{model} \end{cases}$

$\Rightarrow G^2(M_0) = -2(L_0 - L_S) \sim X^2_{df(0)}$ ; $G^2(M_1) = -2(L_1 - L_S) \sim X^2_{df(1)}$

3. Reject $H_0$ if $\Delta G^2 > X^2_{df,\alpha}$

# Logistic Regression

e. g. 螃蟹追求者

$\begin{cases} H_0: \text{logit } \pi(x) = \alpha & (\text{model } 0) \\ H_1: \text{logit } \pi(x) = \alpha + \beta x & (\text{model } 1) \end{cases}$

$\left.\begin{matrix} G^2(M_0) = 34.0 & df(0) = 7 \\ G^2(M_1) = 6.0 & df(1) = 6 \end{matrix}\right\} \Rightarrow \Delta G^2 = G^2(M_0|M_1)$

$= G^2(M_0) - G^2(M_1) = 34.0 - 6.0 = 28 > X^2_{1,0.05} = 3.84$

$df = 7 - 6 = 1 \quad \therefore \text{Reject } H_0$

# Logistic Regression

※Residuals for logit models

Let $y_i$ : the number of "successes" for $n_i$ trials at the i-th setting of the explanatory variables

$\hat{\pi}_i$ : the predicted probability of success for the model fit

$n_i\hat{\pi}_i$ : the fitted number of successes

· The Pearson residual $e_i = \frac{y_i - n_i\hat{\pi}_i}{\sqrt{n_i\hat{\pi}_i(1-\hat{\pi}_i)}} \sim N(0,1)$ $|e_i| > 2$ indicate possible lack of fit

(if the number of model parameters is small compared to the number of sample logits)

· The Pearson statistic $X^2 \cong \sum e_i^2 \sim X_{df}^2$

· 當 $n_i = 1$ 時， $y_i$ 不是 1 就是 0，則 $e_i$ 只有兩個可能值。

在分析時，須特別注意 "extreme value" 及 "single residual" 的問題。

# Logistic Regression

※Diagnostic measure of influence

(1) Leverage (槓桿點) : the i-th diagonal of hat-matrix, 記作 $h_i$

where the hat-matrix is $P_X = X(X'X)^{-1}X'$

i. e. $h_i$ is the i-th diagonal element of $P_X$

· The greater an observation's leverage, the greater its potential influence.

# Logistic Regression

※Diagnostic measure of influence

**(2) Influence measures for each observation**

1. Df beta $= \frac{\hat{\beta}-\hat{\beta}^{(i)}}{s.e.(\hat{\beta}-\hat{\beta}^{(i)})}$, $\hat{\beta}^{(i)}$ : the parameter estimate when the i-th observation is deleted
2. A measure of change in a joint confidence interval for the parameter produced by deleting the observation.
3. The change in $X^2$ or $G^2$ goodness-of-fit statistics when the observation is deleted.

· For each measure, the larger the value, the greater the observation's influence.

· The adjusted residual : $\frac{y_i-\hat{y}_i}{\sqrt{Var(y_i-\hat{y}_i)}} = \frac{y_i-n\cdot\hat{\pi}_i}{\hat{\sigma}\sqrt{1-h_i}} = \frac{y_i-\hat{y}_i}{\sqrt{n_i\cdot\pi_i\cdot(1-\pi_i)\cdot(1-h_i)}}$

where $h_i$ is the i-th diagonal element of hat-matrix.